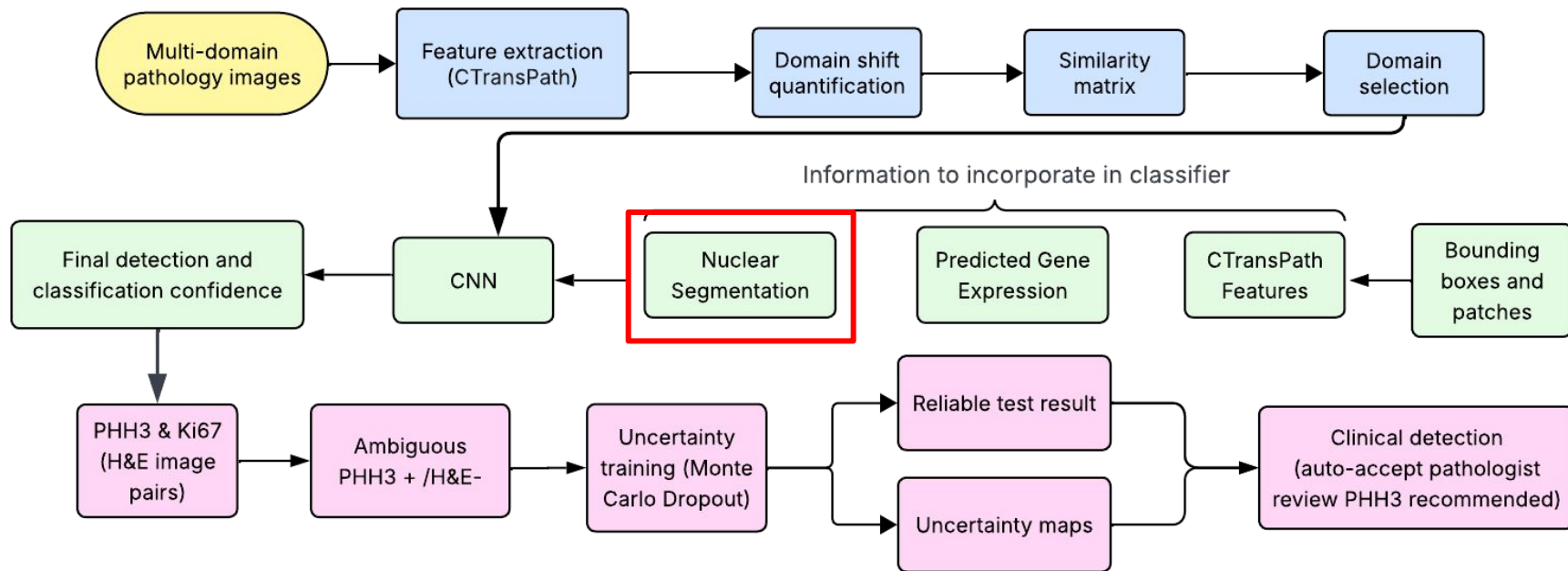




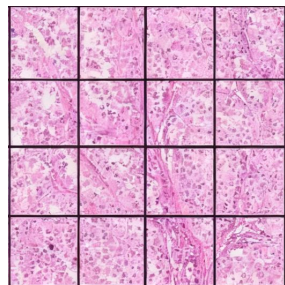
Training Update

Lillian Lam | Naina Misra | Yang Cao | 3/5/2026

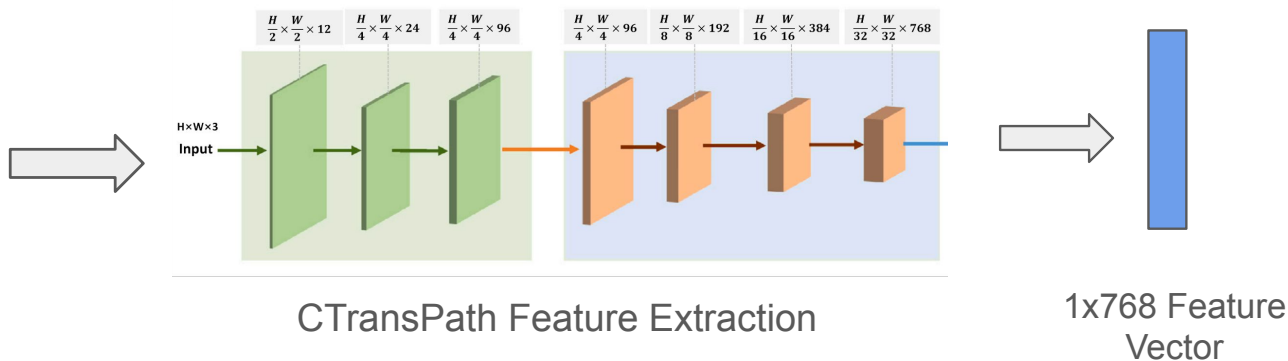
Pipeline



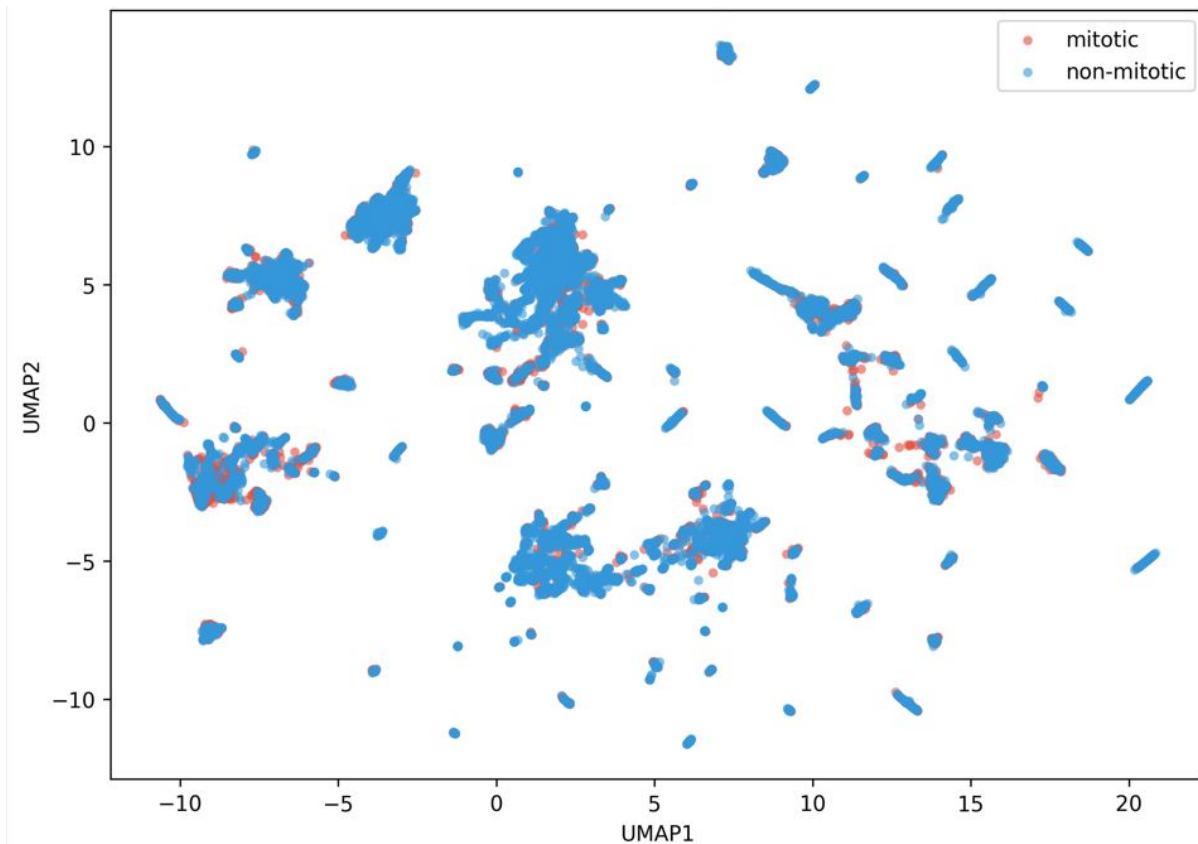
Recap



224x224 Patch
Extraction around
annotated bbox

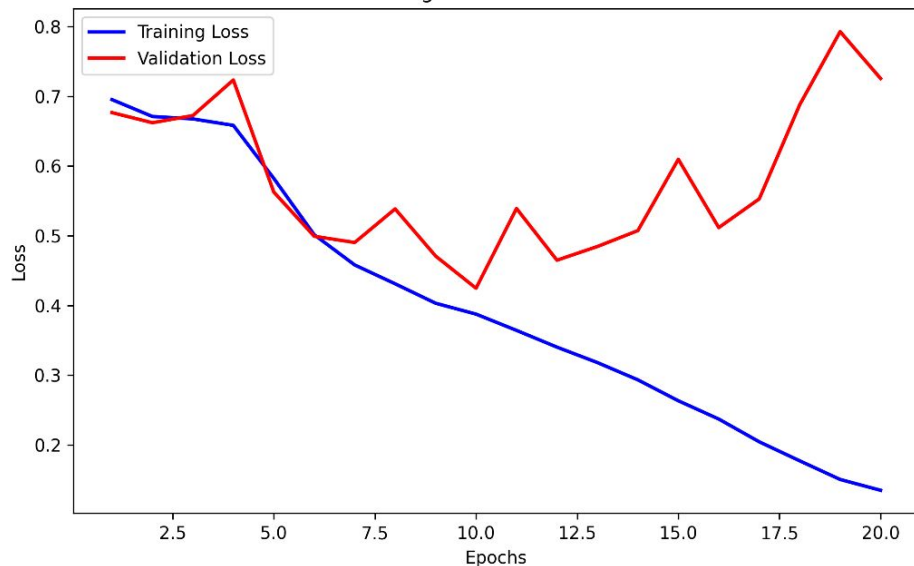


Mitotic and Non-Mitotic Separation in Ctranspath

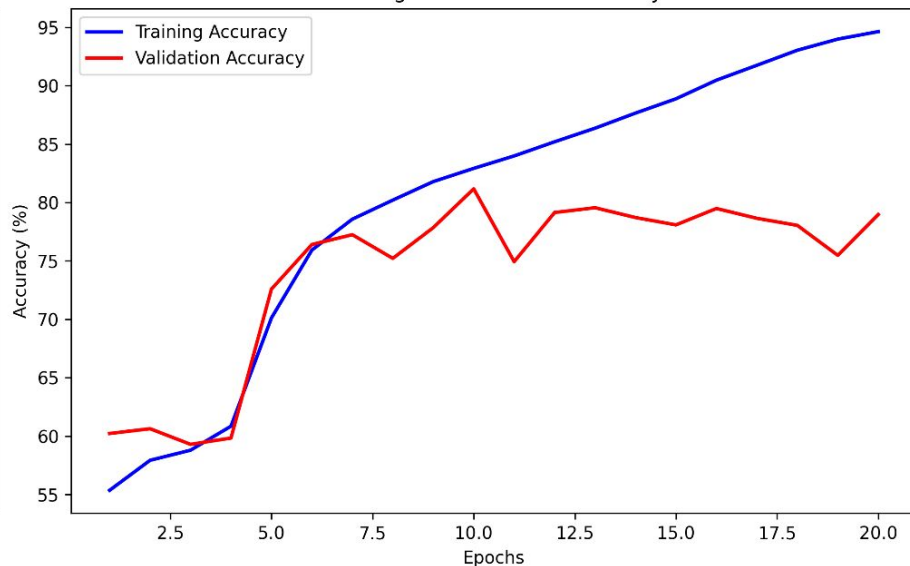


Control Results

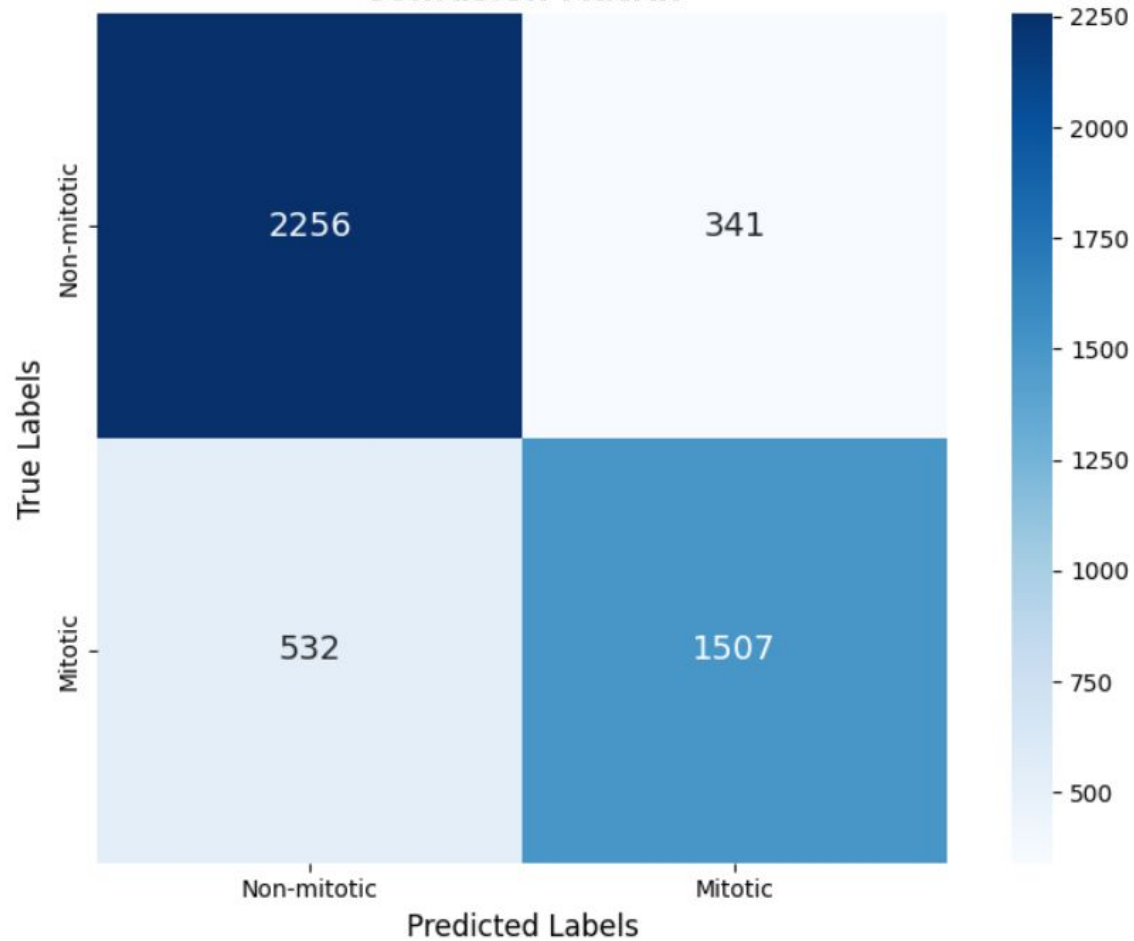
Training and Validation Loss



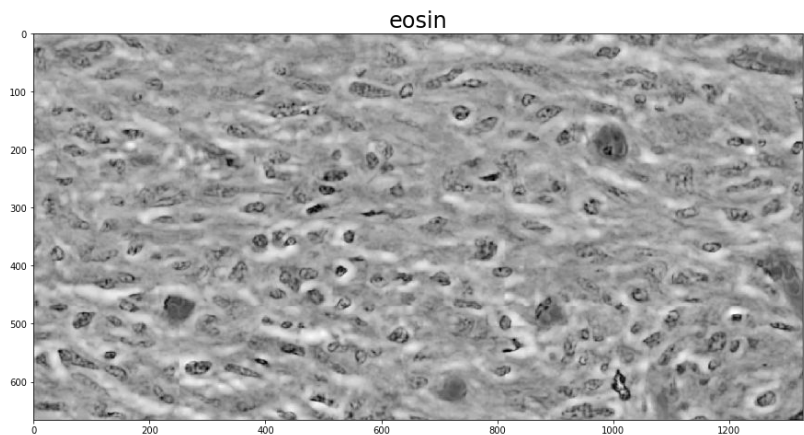
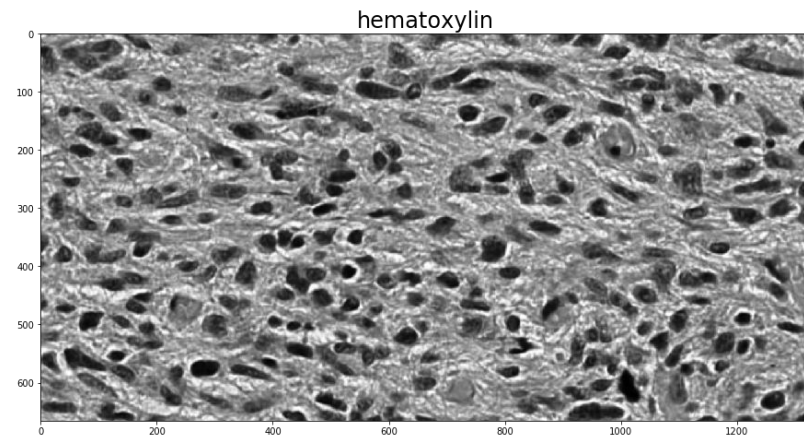
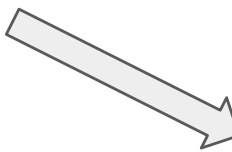
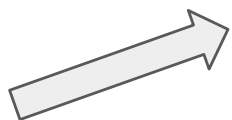
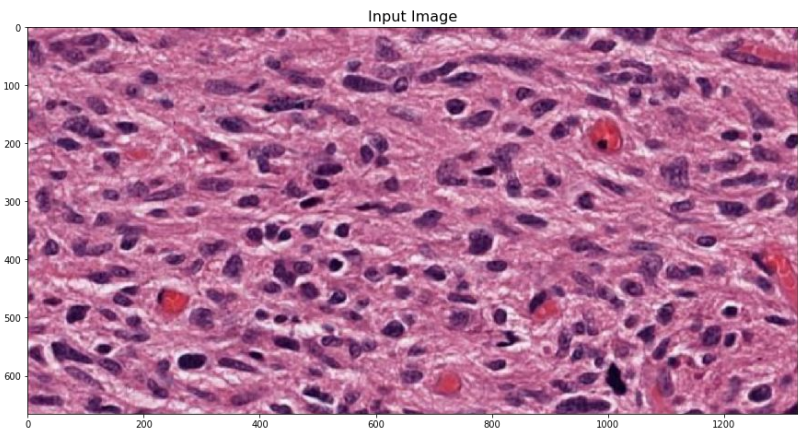
Training and Validation Accuracy



Confusion Matrix



HistomicsTK Color Deconvolution



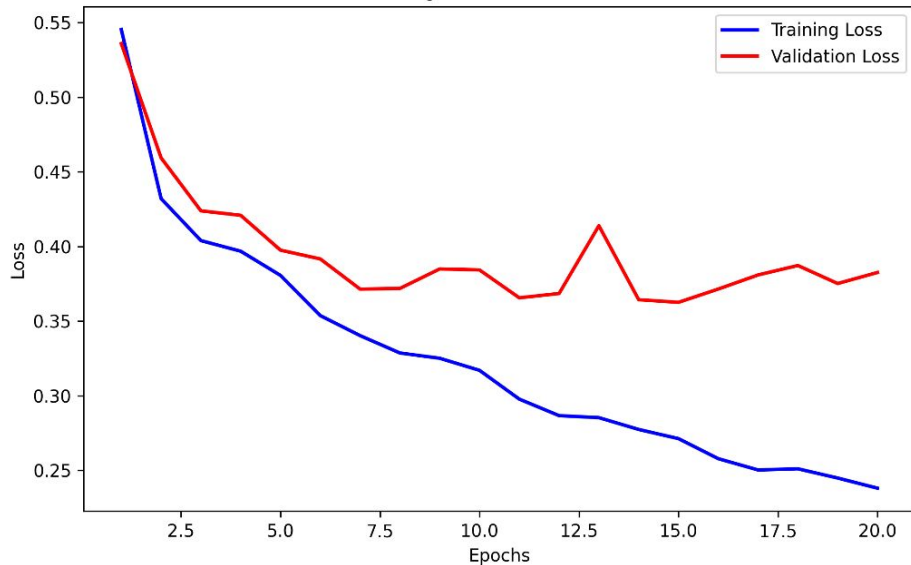
How it works

Optical density: measure of how much light gets absorbed when passing through a material.

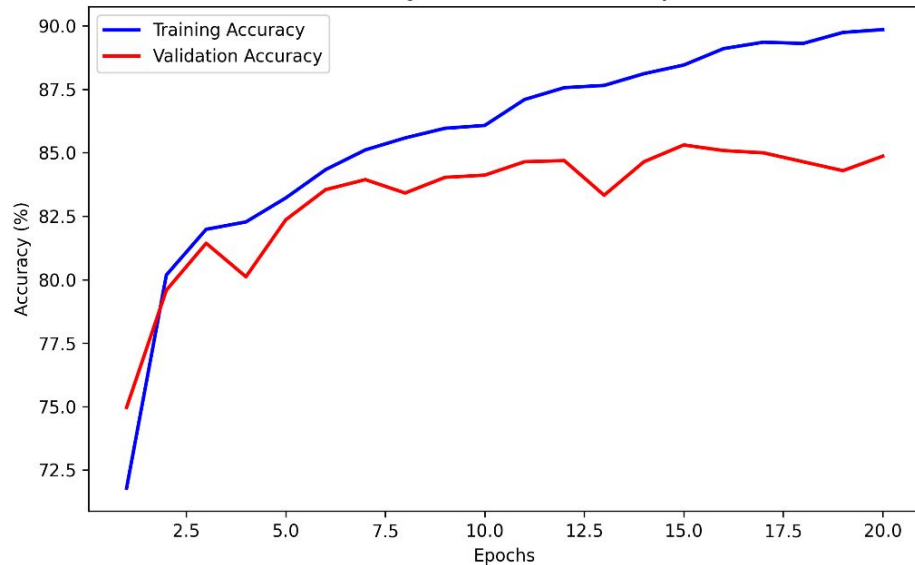
Stain Amount	What You See	Optical Density
No stain	Very bright/white	Low (near 0)
Light stain	Light pink/blue	Medium
Heavy stain	Dark purple	High

Color Deconvolution Results

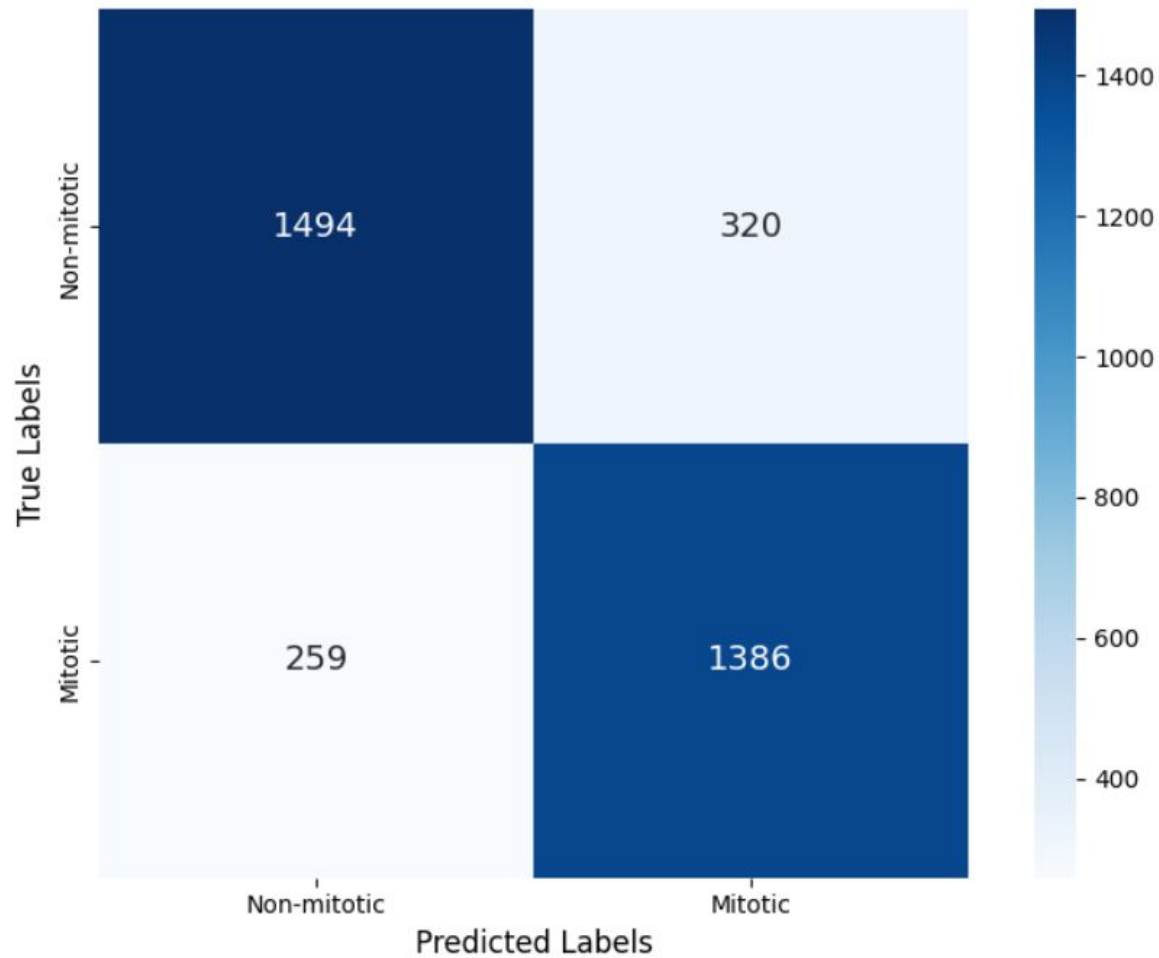
Training and Validation Loss



Training and Validation Accuracy

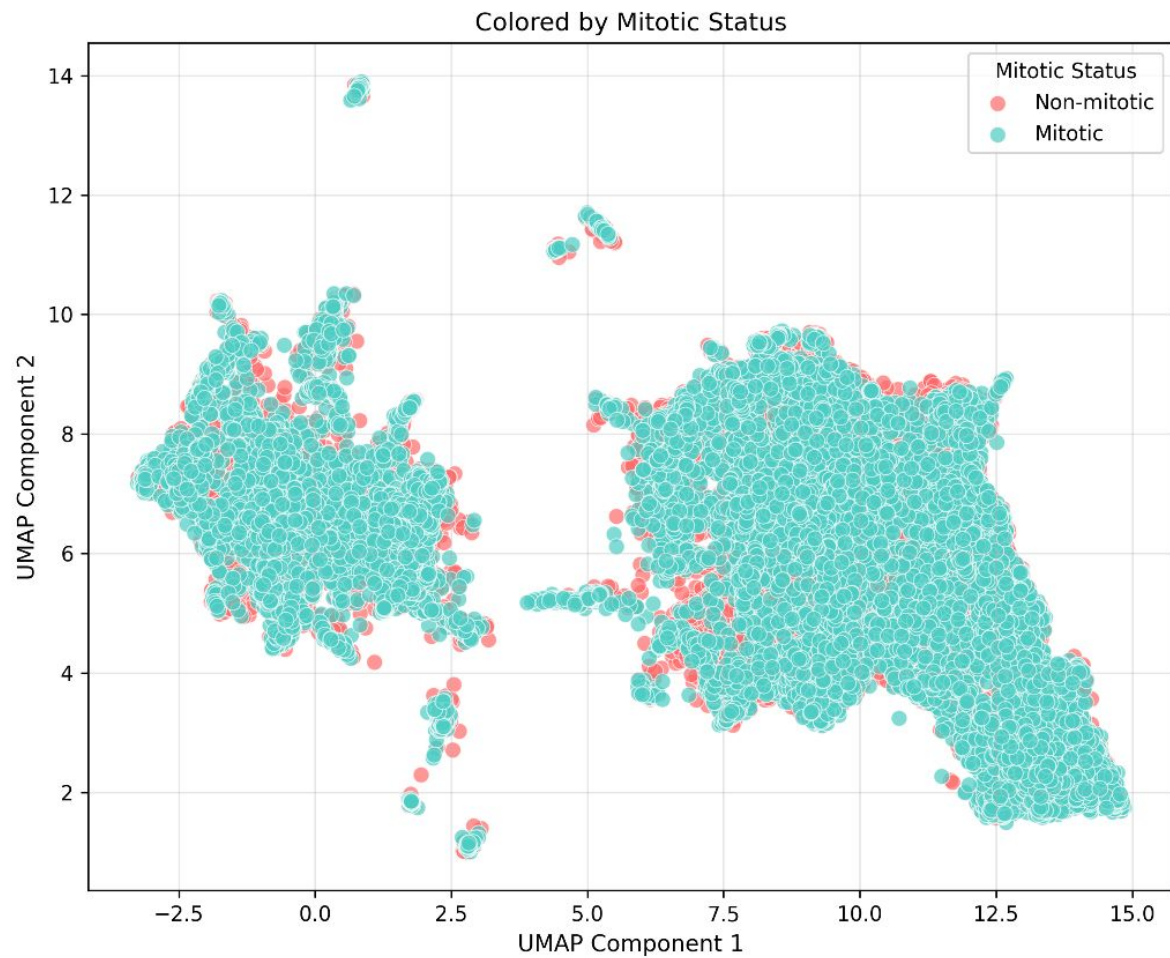


Confusion Matrix



Comparison

	Control	Color Deconvolution
Mitotic F1	0.78	0.83
Non-Mitotic F1	0.84	0.84
Mitotic Recall	0.74	0.84
Mitotic Precision	0.82	0.81



Idea (Adversarial Penalty)

Feature Extractor: learns image representations

Mitosis Classifier: predicts mitotic vs non-mitotic (primary task)

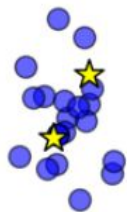
Domain Classifier: tries to predict which slide the patch came from (adversarial task)

Goal:

- Feature extractor tries to fool the domain classifier
- Domain classifier tries to identify the slide source
- Layer penalizes features that reveal slide identity

Without Penalty

Slide A



Slide B



Clustered by Slide

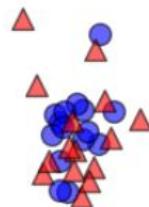
● Slide A

▲ Slide B

★ Mitotic

With Penalty

Non-Mitotic



Mitotic



Clustered by Biology

Next Steps

- Try different HE to RNA-seq
- Fine-Tune hyperparameters
- Ki67 predictions with IHC stains
- Uncertainty quantification

Thank you!